

# Two Affirmative Answers for Continuous Operator Families

Literature-implied status note for arXiv:2010.07040

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## Status and source questions

**Status: literature-implied answer (both questions, affirmative).**

After Theorem 1.5, Berkani asks the following two questions [1, p. 4].

1. Given  $S \in C(\mathbb{X}, L(X)/K(X))$ , must there be  $T \in C(\mathbb{X}, L(X))$  whose pointwise Calkin image is  $S$ ?
2. If  $H$  is a Hilbert space and  $\mathcal{K} \in C(\mathbb{X}, K(H))$ , is  $\mathcal{K}$  a uniform norm limit of continuous finite-rank families?

Here  $\mathbb{X}$  is compact and all function spaces carry the supremum norm.

## Question 1: continuous lifting

Let

$$q : L(X) \longrightarrow L(X)/K(X)$$

be the quotient map. It is a continuous linear surjection between Banach spaces. The classical Bartle–Graves theorem says that every such map has a continuous, positively homogeneous right inverse [2, Theorem 5.1, p. 14]; thus there is a continuous  $s : L(X)/K(X) \rightarrow L(X)$  with  $q \circ s = \text{id}$ .

For the given family  $S$ , define

$$T_x = s(S_x) \quad (x \in \mathbb{X}).$$

Then  $T = s \circ S$  is continuous and  $q(T_x) = S_x$  for every  $x$ . Since  $\mathbb{X}$  is compact,  $T$  is automatically bounded and belongs to the source's Banach algebra  $C(\mathbb{X}, L(X))$ . Therefore the natural map

$$C(\mathbb{X}, L(X))/C(\mathbb{X}, K(X)) \longrightarrow C(\mathbb{X}, L(X)/K(X))$$

is surjective. In fact, the lifting conclusion itself does not require compactness of the parameter space.

## Question 2: one compression for the whole family

Let

$$\mathcal{C}_0 = \{\mathcal{K}_x : x \in \mathbb{X}\} \subset K(H).$$

This is compact because it is the continuous image of the compact space  $\mathbb{X}$ . Fix  $\varepsilon > 0$ . Choose a finite  $\varepsilon/4$ -net  $K^{(1)}, \dots, K^{(m)}$  in  $\mathcal{C}_0$ . Since finite-rank operators are norm dense in  $K(H)$ , choose finite-rank  $F_j$  with

$$\|K^{(j)} - F_j\| < \varepsilon/4.$$

Let  $E$  be the finite-dimensional span of all  $\text{ran } F_j$  and  $\text{ran } F_j^*$ , and let  $P$  be the orthogonal projection onto  $E$ . The two range inclusions give  $PF_j = F_j$  and  $F_jP = F_j$ , hence  $PF_jP = F_j$ .

For any  $x$ , select  $j$  with  $\|\mathcal{K}_x - K^{(j)}\| < \varepsilon/4$ . Then

$$\|\mathcal{K}_x - F_j\| < \varepsilon/2$$

and, because  $P$  is contractive,

$$\|\mathcal{K}_x - P\mathcal{K}_xP\| \leq \|\mathcal{K}_x - F_j\| + \|P(F_j - \mathcal{K}_x)P\| < \varepsilon.$$

The family  $x \mapsto P\mathcal{K}_xP$  is norm-continuous and every member has range in  $E$ , so it is a continuous finite-rank family. Taking  $\varepsilon = 1/n$  produces the requested uniformly convergent sequence. No separability assumption on  $H$  is needed.

## Provenance and scope

The first answer is a direct specialization of the Bartle–Graves theorem, which long predates the source question. The second is the standard density of finite-rank operators in  $K(H)$  made uniform over a compact subset by a single finite-dimensional compression. The relation to Berkani’s two questions is agent-identified: the supporting authors do not claim to answer them.

A bounded search through August 11, 2026 used the exact arXiv id and title, the question wording, continuous Calkin lift terms, Bartle–Graves, and continuous finite-rank-family terms. It found the source, the supporting selection theorem, and general uses of continuous Calkin sections, but no later paper explicitly presenting itself as a resolution of these two questions.

## References

- [1] M. Berkani, *Index of continuous families of bounded linear operators in Banach spaces and application*, arXiv:2010.07040 (2020), <https://arxiv.org/abs/2010.07040>.
- [2] M. Messerschmidt, *A pointwise Lipschitz selection theorem*, arXiv:1611.08435; see Theorem 5.1 for the classical Bartle–Graves theorem, <https://arxiv.org/abs/1611.08435>.
- [3] R. G. Bartle and L. M. Graves, *Mappings between function spaces*, Trans. Amer. Math. Soc. **72** (1952), 400–413.